The Box Principle — problem 4.53

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Prove that there is a power of two, which begins with 6 nines, that is, there are positive integers n, k such that

$$999999 \times 10^k < 2^n < 10^{k+6}, \qquad k + \log 999999 < n \log 2 < k + 6.$$

Hint. Here $\epsilon = 6 - \log 999999$ and the step length is $\alpha = \log 2$. Similarly, one can show that, for irrational $\log a$, there is a power of a which begins with any prescribed digit sequence.